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Effect of Propeller Incidence Angle on Wing Embedded Propeller Configuration in Forward Flight

George Gogidze[1] and Jielong Cai[2]

University of Dayton, Dayton, Ohio, 45419, United States

Sidaard Gunasekaran[3]

University of Dayton, Dayton, Ohio, 45419, United States

We continue our study on the effect of a propeller embedded in the wing under various propeller incidence angles. The current study focuses on the performance of the wing and the wing-embedded propulsion as one system. Three semi-span AR2 wings with different wing hole shapes at mid-semi-span were experimentally investigated with and without the propeller installed. Minor differences were found between different wing hole shapes without the installation of the propeller in the wing. With the propeller installed, the difference between wing hole shapes becomes evident. The circular wing hole has the best performance among all cases tested. An empirical model is proposed to determine the lift coefficient of the propeller-in-wing system as a function of standalone propeller and wing performance. However, the modeling of the forward thrust from the system posed challenges.

Nomenclature

AR	=	Aspect Ratio
C_D	=	Drag Coefficient, $C_D = \frac{D}{qS_{Ref}}$
C_L	=	Lift Coefficient, $C_D = \frac{L}{qS_{Ref}}$
C_M	=	Moment Coefficient, $C_D = \frac{M}{qS_{Ref}}$
C_{Tp}	=	Forward Thrust Coefficient, $C_{TP} = \frac{T_P}{qS_{Ref}}$
D	=	Drag force, (N)
d	=	Distance, (m)
J	=	advanced ratio, $J = \frac{V_\infty}{nD}$
L	=	Lift force, (N)
L_P	=	Propeller lift force, (N)
M	=	Pitching Moment, $(N.m)$
n	=	Propeller Rotational Speed, (RPS)
P	=	Power, (W)
q	=	Dynamic Pressure, (Pa)
Re	=	Reynolds Number
RPS	=	Rotations Per Second
S_{ref}	=	Reference area, (m^2)
T	=	Thrust Force, (N)
T_P	=	Propeller Forward Thrust, (N)
V_∞	=	Freestream Velocity, (m/s)
α	=	Angle of Attack, (Deg)
ρ	=	Air density, (kg/m^3)
θ	=	Incidence angle, (Deg)

I. Introduction

Most modern personal air vehicles (PAV) and unmanned air vehicles (UAV) design integrate vertical takeoff and landing (VTOL) capabilities with fixed-pitch multi-rotors, distributed electric propulsion, and wing-embedded propulsion (WEP) system [1]. The wing-embedded propulsion systems have been considered since the 1960s in designs such as the Ryan XV5 by Ryan Aeronautical shown in Figure 1 [2]. The WEP configurations were found easy to maintain and operate but was difficult to control especially during the aircraft's transition and higher angle of attack operation.[2]. More recent design concepts include AgustaWestland Project Zero (Figure 1) featuring a WEP system

[1] Undergraduate Student, Department of Mechanical and Aerospace Engineering, AIAA Member. gogidzeg1@udayton.edu
[2] Graduate Student, Department of Mechanical and Aerospace Engineering, AIAA Member. caij03@udayton.edu
[3] Assistant Professor, Department of Mechanical and Aerospace Engineering, AIAA Member. gunasekarans1@udayton.edu

capable of rotating the propeller to provide thrust vectoring. However, it only managed to perform vertical takeoff and landing, and the capability of transition to forward flight is never achieved [4].

In our prior work [5-6], a propeller was embedded in a wing with two different wing hole shapes, square and circular, located at the mid-span. The effect of the wing on the propeller was quantified through the changes in propeller performance. When placing the propeller in the hole, a consistent loss in the propeller forward thrust is observed between a propeller incidence angle, θ, of 55 and 90 degrees. This loss increases with the increment in advance ratio, J. On the other hand, a small reduction in the propeller lift (normal to the freestream) generation was observed at $\theta > 60\,°$. A vane system was then placed at various angles beneath the propeller to provide thrust vectoring for the transition to forward flight application. A significant loss in the forward thrust was observed at $\theta = 75°$ under all vane deflection angles tested when compared to the standalone propeller while the loss in the lift generation was relatively small. At $J < 0.2$, the vane system was able to provide both pitch-up and pitch-down moments for transition stability by deflecting the vane angle. For the wing hole shape, on average, the circular wing was found to be 10.5% more efficient on propeller thrust vectoring performance when compared to the square hole shape. Our previous study provided significant insight into the effect of embedding a propeller inside a wing on the propeller, the overall wing performance of the wing-embedded propulsion system was not investigated.

Ref [7] studies the effect of holes on the wing (represented by circular holes on the wing) performance with a NACA 64-412 airfoil. The study found that damage to the leading edge at low angles of attack created a significant decrement in lift generation while trailing edge damage proved to be least significant. Additionally, the damage near the quarter chord and half chord proved to be the most significant in worsening the lift performance due to drastic changes in pressure distribution. Ref. [1-3] examined different WEP designs including large fans integrated into the wing for lift production and smaller fans integrated which provide roll and pitch moment. In most cases, placing a rotor in the wing resulted in a positive effect on lift generation. However, placing the fan closer to the leading edge of the wing caused a reduction of the flow speed over the upper wing surface, resulting in significant lift penalties.

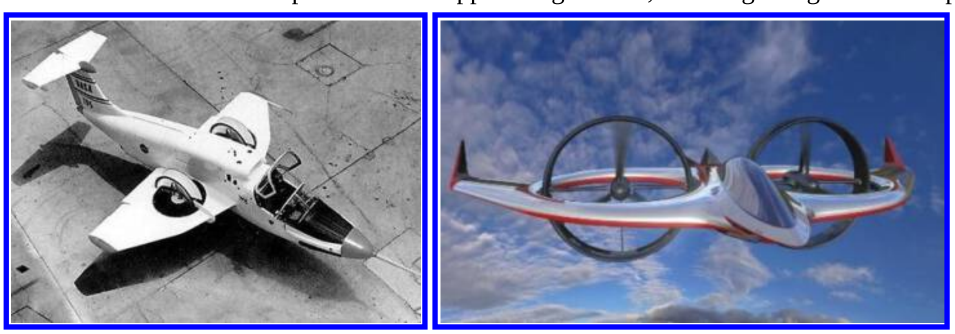

Figure 1. Ryan XV-5 (left) [2] and AgustaWestland Project Zero (right) [4]

Despite the increase in WEP configurations in modern UAVs and PAVs, the effect of propeller incidence on static and transition performance remains unexplored [8-10]. Thus, the objective of the current study is 1) to explore the effect of propeller incidence on the overall WEP system performance, 2) to characterize the changes in wing performance due to propeller presence in the wing and incidence angles, 3) to investigate the influence of hole shapes that houses the propeller in the wing, and 4) to develop a prediction tool that would allow lift prediction of the WEP system using the performance of the lift generated by a standalone wing and the thrust generated by a standalone propeller.

II. Experimental setup

All experiments were conducted in the University of Dayton's Low Speed Wind Tunnel (UD-LSWT) in its open jet configuration. The tunnel has a 16:1 contraction ratio. 6 anti-turbulence screens, and a 76.2 cm by 76.2 cm open jet test section. The effective length of the test section is 182.9 cm. A 111.8 cm by 111.8 cm collector accepts the expanded air on its return to the diffuser. A pitot tube is connected to a TSI T600 Micromanometer which was used to measure the freestream velocity during the experiment. The experimental setup that was conducted for the solo wing studies as well as the combined system of the propeller and wing system is shown in Figure 2.

A. Standalone Wing Test Setup

Three different wing hole types, with a propeller installed and operating (with propeller) and without a propeller installed (without propeller), along with the baseline wing with no holes are tested in the experiment, shown in Figure

3. All wings were designed using the Eppler 479 airfoil and has a span of 0.254 m and a chord of 0.127 m. The reference areas for the wing with holes are lower than the baseline wing without any holes. The wings were mounted to a rotary stage and an ATI Gamma F/T [11] balance assembly. A splitter plate size of 0.91m x 1.83m was used to obtain a half-span configuration of the models tested. The tunnel speed for the no-propeller wing cases was set at 20 m/s which corresponds to a chord-based Re of 293,000. The angle of attack was varied from -20 to 20 degrees with 1° increment.

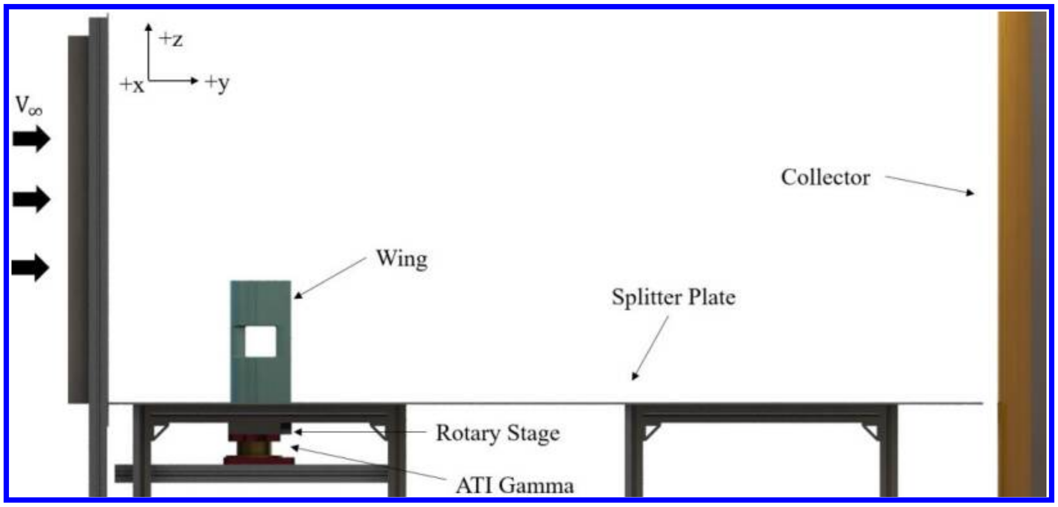

Figure 2. Rendered CAD schematic of UD-LSWT experimental setup

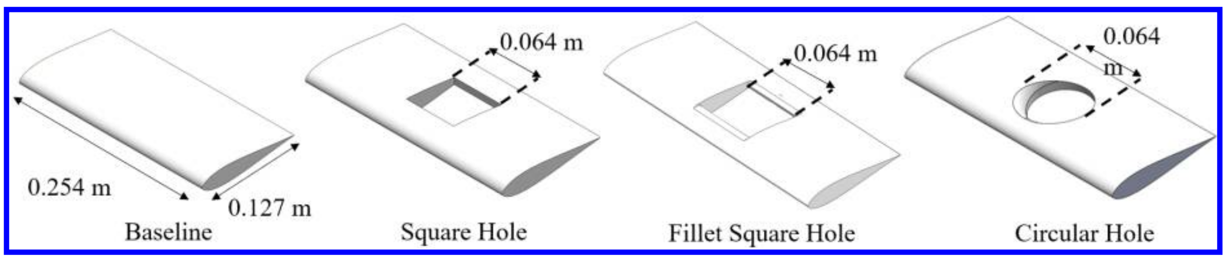

Figure 3. Schematic of the Different Wing Shapes Studied

B. Standalone Propeller Test Setup

For the wing-embedded propulsion configuration experiment, a T-motor T-2004 brushless motor [12] and an eight-blade 2.5x1.5 (63mm diameter and 38 mm pitch) propeller were selected, which scaled to the overall hole/wing ratio when compared to our previous study [5-6]. The single propeller performance was tested to set up a baseline for the propeller. The propeller rotational speeds were set at 500 and 625 RPS, with freestream velocities ranging from 0 to 20 m/s, resulting in an advance ratio between 0 and 0.65. The propeller was tested at different incidence angles between 45 to 90 degrees. Figure 4 depicts the motor's setup in the UD-LSWT. The motor is mounted on a sensor interface that is 2.5 times the propeller diameter away from the ATI-Mini-40 F/T balance [13] to avoid any interference. The sensor is then mounted on the sensor mounting angle interface to connect to the aluminum frame which was secured to the laboratory floor. The sensor mounting angle interface was changed to change the propeller incidence angle which kept the propeller thrust, and torque in line with the sensor's balance axis.

C. Propeller-in-Wing Test Setup

The test matrix of the different Propeller-in-Wing test cases investigated is shown in Table 1. The wings were tested at a velocity range of 10 m/s to 20 m/s, which is an equivalent of the propeller advance ratio of 0.3 and 0.6 sweeping through angles of attack ranging from -20 to 20 degrees angle of attack with an angle increment of 1° at each velocity The schematic of the propeller installment on the wing is shown in Figure 5. The propeller and motor were mounted on a motor interface which is secured to the wing body. To change the propeller incidence angle, this mounting interface is replaced. The interface is designed to keep the center of the propeller hub at the upper surface of the wing which is consistent with the condition in our previous study [5-6].

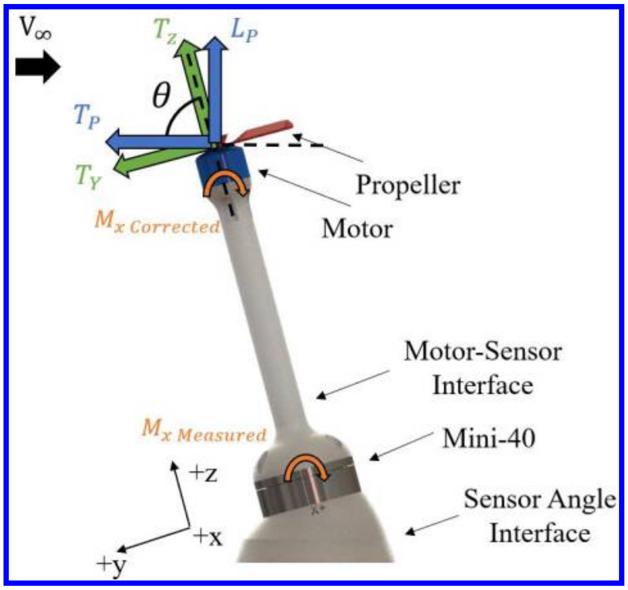

Figure 4. Schematic showing the single propeller experimental setup

Table 1. Test Matrix of the Propeller-in-Wing Experiment

Wing Configuration	Reference Area(m^2)	Wing Reynolds Number	Wing angle of attack (degrees)	Freestream Velocity (m/s)	Propeller Rotational Speed (RPS)	Propeller incidence angle(degrees)
Baseline	0.03226	290,000	-25 to 25	15	N/A	N/A
Circular Hole	0.02932	84,000 to 168,000	-20 to 20	0,10, 15, 20	500, 625	60, 75, 90
Square Hole	0.02856					
Fillet Square Hole	0.02856					

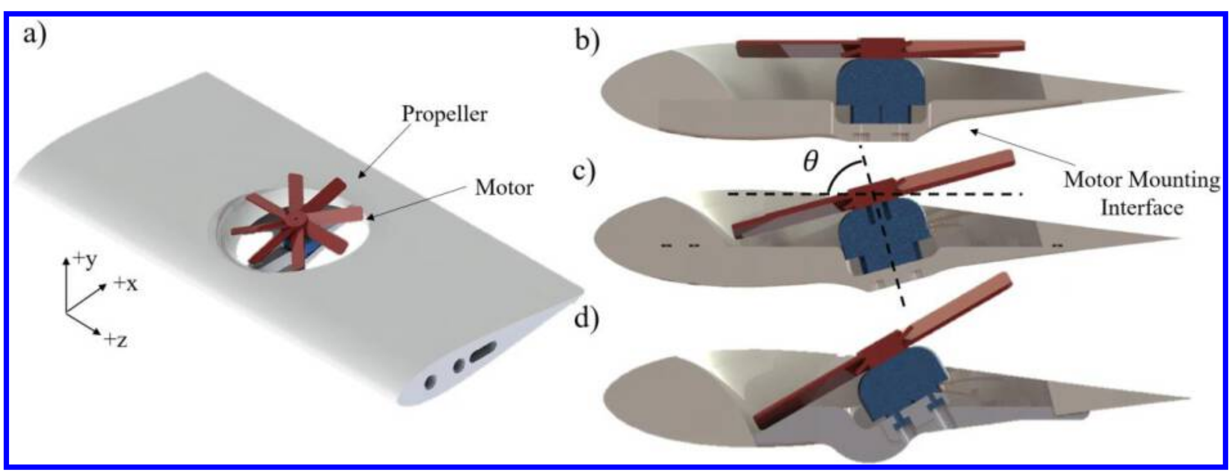

Figure 5. Schematic showing the wing-embedded propulsion configuration with a) isometric view b) $\theta = 90°$, c) $\theta = 75°$ and d) $\theta = 60°$

D. Numerical Setup with FlightStream®

FlightStream® [14] was used to numerically investigate the standalone wing with different hole shapes. This provides a fast estimation of the performance of the standalone wing which can be used to predict the performance of the WEP system later on. FlightStream® contains two main types of potential flow solvers: pressure-based solver and vorticity solver. The pressure-based solver is used in the present study, which utilizes the pressure fields on the model geometry to determine the aerodynamic loads. A combination of source and sink panels along with doublet panels on the trailing edge is used to provide a solution for pressure-based potential flow solvers.

The surface mesh of the wing and the splitter plate in FlightStream® is shown in Figure 6. The wing platform has a structured mesh with 80 node points spanwise and 60 node points chordwise on both upper and lower surfaces. The chordwise mesh also has a dual side growth rate of 1.05 to better resolve the leading and trailing edge geometry. An unstructured mesh is used for the circular hole to better represent the complex geometry. Moreover, the splitter plate of the same size as the one used wind tunnel experiment is included in the model to better represent the actual experimental condition. The splitter plate has 40 node points from side to side and 60 node points, with a dual-side growth rate of 1.05 downstream.

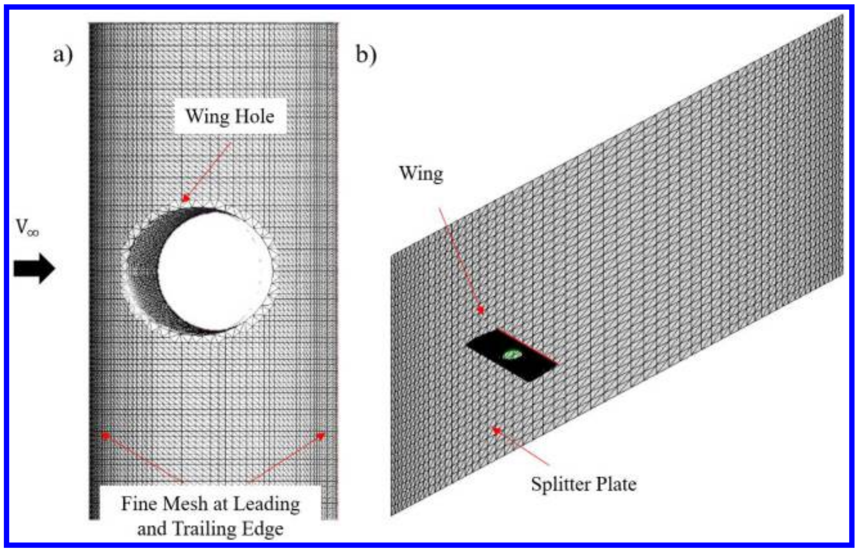

Figure 6: Surface mesh in Flightstream for a) circular hole wing and actuator propeller and b) wing and splitter plate

III. Experimental Results

The experimental results will be discussed in sub-sections. The first sub-section discusses the performance results from standalone wings with different hole shapes without the propeller. The second sub-section focuses on the performance of the standalone propeller. Then, the performance results from the different propeller-in-wing cases are discussed. The final sub-section compares the performance between the standalone propeller and wing cases and the wing-embedded propulsion systems.

A. Standalone Wing Performance

The coefficient of lift and drag of the standalone wing cases measured from the wind tunnel experiment and estimated from FlightStream® are shown in Figure 7. The experimental result is shown as scatter points while the FlightStream® results are shown as dashed lines. The surface area of each wing shown in Table 1 was used to normalize the lift and drag forces to obtain lift and drag coefficients. For the experimental results, the lift coefficient of all cases with the holes has a lower lift curve slope than the baseline as expected. The effect of the different wing hole shapes on its lift coefficient is similar. Based on the McCormick formula [15], the effective AR calculated based on the lift curve slope for the baseline wing is 2.94, adding the hole reduces the effective AR down to 2.1. On the other hand, the wings with holes have an overall drag higher than that of the baseline for all angles of attack. The square wing hole has the highest C_D among all cases postulated due to large flow separation and wake shedding from its sharp edges. Adding the fillet helps reduce C_D for the square hole at all α tested. In general, similar to the lift force, the difference in drag force for different wing hole shapes is also very small. On the other hand, FlightStream® was able to predict the C_L reasonably well for all cases within the stall α. The C_D by FlightStream® for the baseline wing also agreed well with the experimental result before the stall. However, the C_D prediction only agreed with experimental data within $-7 < \alpha < 7$ for all cases with holes. As expected, the solver was not able to predict the parasitic drag increment due to flow separation around the hole region.

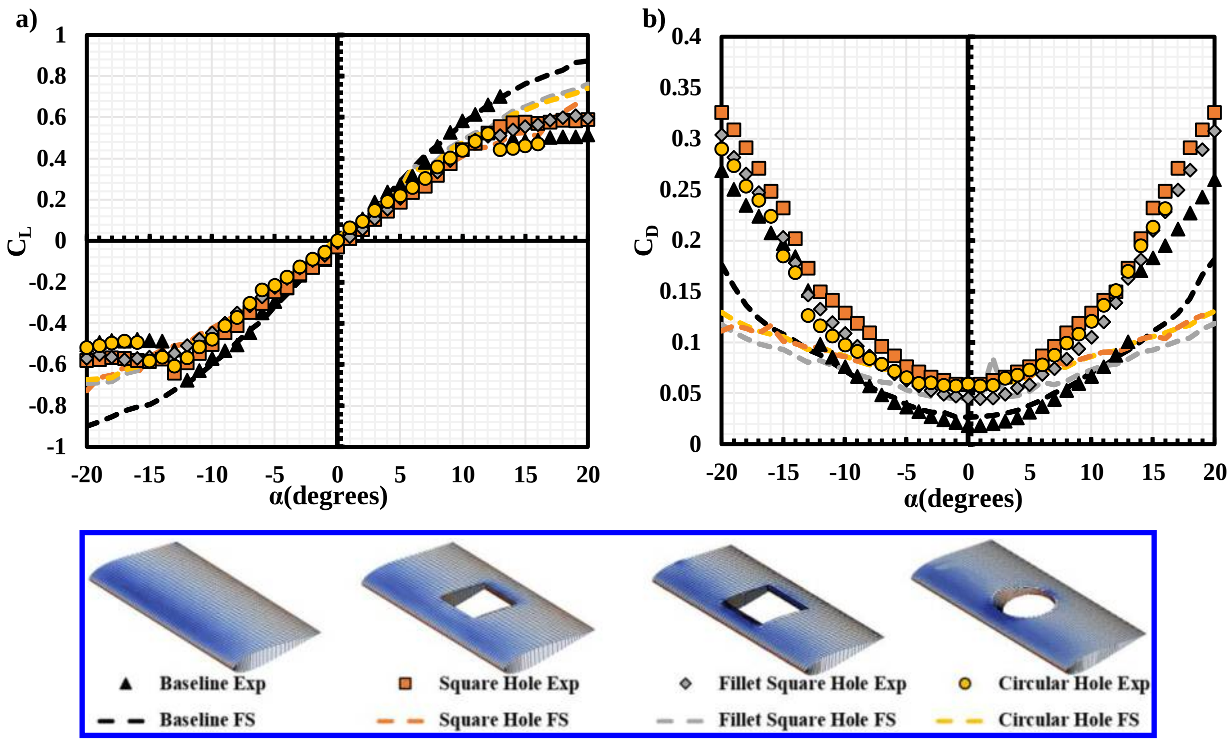

Figure 7: a) Lift coefficient and b) drag coefficient vs. α for the standalone wing cases

B. Standalone Propeller Performance

The dimensional axial thrust and power measured from the standalone propeller are shown in Figure 8. The result is shown in terms of the dimensional values as the conventional non-dimensional term, thrust coefficient, and power coefficient, cannot be directly compared to the non-dimensional term for the wing performance, which is the lift coefficient and drag coefficient. Two different rotational speeds at four different incidence angles were tested. The thrust force is the axial force about the F/T balance Z-axis while the power is calculated based on the torque measured about the F/T balance Z-axis.

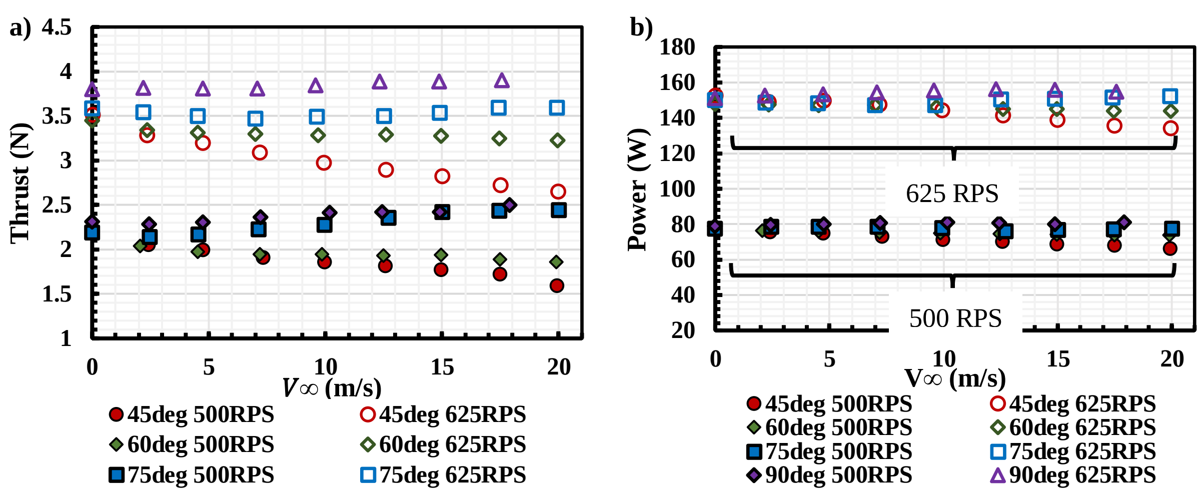

Figure 8. a) Propeller axial thrust and b) power v.s freestream velocity

At the same incidence angle (θ), the propeller running at a higher rotational speed produces higher thrust and consumes higher power, as expected. For the same rotational speed, at $\theta > 75°$, the measured thrust and power increase slightly with the increment in freestream velocity, V_∞, which represents the advance ratio. This trend is reversed for $\theta < 75°$. It is also worth noticing that the difference in the power consumption at low freestream velocity is negligible until higher V_∞. A similar trend for propeller performance under a high incidence angle is also found in other studies [16-19].

To better compare the propeller performance in the WEP system, the propeller vertical lift (L_P) and forward thrust (T_P) are calculated which represents the propeller's performance with respect to the wing orientation, and can be expressed as:

$$\begin{bmatrix} T_p \\ L_P \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} F_Z \\ F_Y \end{bmatrix} \tag{1}$$

Figure 9 shows the calculated $\boldsymbol{L_P}$ and $\boldsymbol{T_P}$ for both RPS cases under different incidence angles at different freestream velocities. At the same $\boldsymbol{\theta}$, the $\boldsymbol{L_P}$ increases slightly with the increment of freestream velocity, indicating the overall ability of the propeller in generating the lift force remains relatively consistent at the same incidence angle. With the increment in $\boldsymbol{\theta}$, the overall magnitude of $\boldsymbol{L_P}$ increases, as expected. The overall magnitude of $\boldsymbol{T_P}$ decreases with the increment in $\boldsymbol{\theta}$ as expected. At $\boldsymbol{\theta < 60°}$, $\boldsymbol{T_P}$ decreases with the increment in $\boldsymbol{V_\infty}$, while this trend reverses for higher $\boldsymbol{\theta}$ cases due to the wake deflection being dominated by the freestream rather than the propeller thrust.

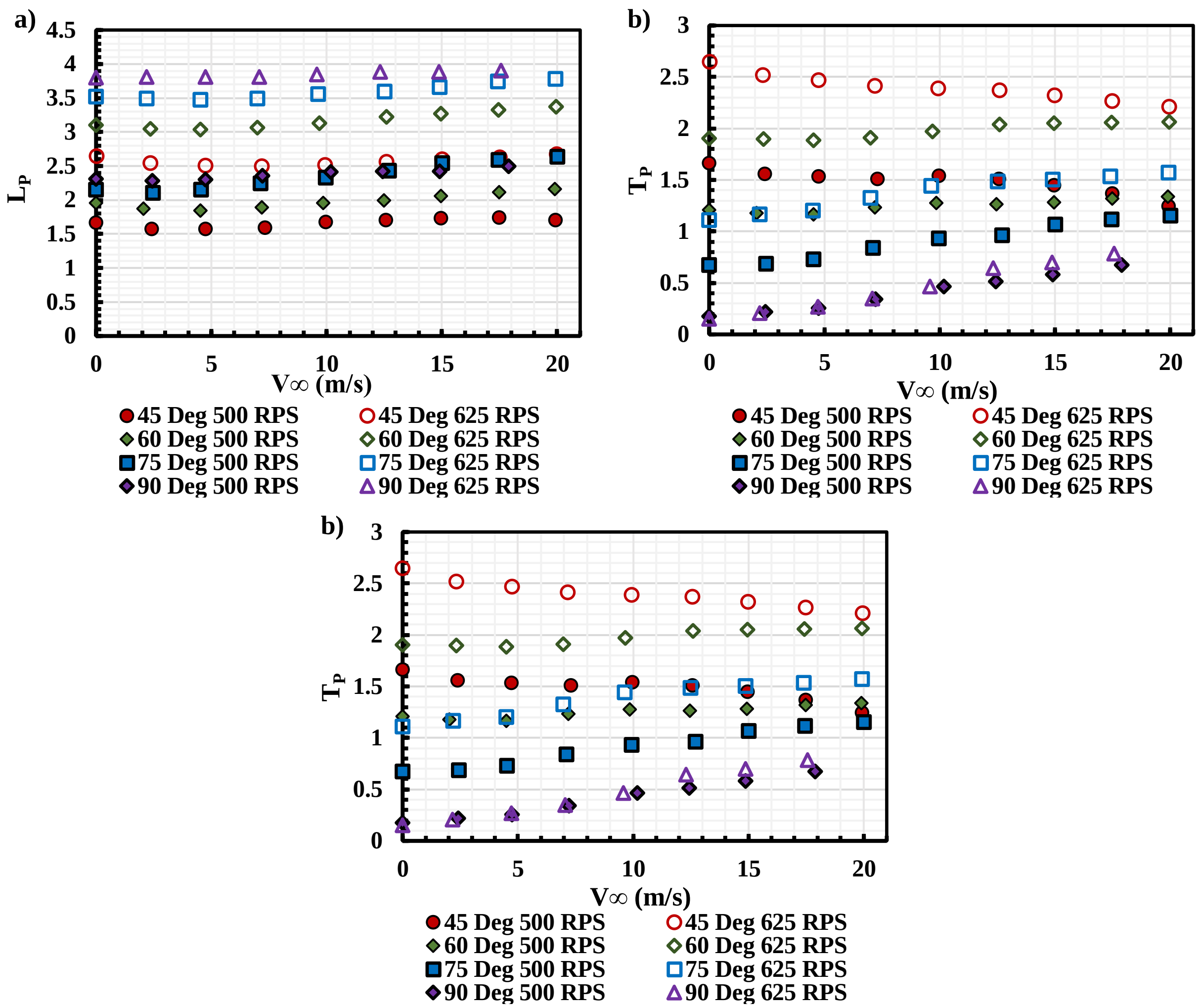

Figure 9. a) L_P and b) T_P under different θ v.s V_∞ at 500 and 625 RPS

Considering the propeller as a lifting surface as a wing, we can normalize L_P and T_P in terms of the lift coefficient (C_{L_P}) and forward thrust coefficient (C_{T_P}) with the conventional method applied on a wing. We can also normalize the propeller's total thrust T_{total} as:

$$C_{L_P} = \frac{L_P}{qS_{prop}} \qquad C_{T_P} = \frac{T_P}{qS_{prop}} \qquad C_{T_{total}} = \frac{T_{total}}{qS_{Prop}} \tag{2}$$

Where q is the dynamic pressure of the freestream and S_{prop} is the disk area of the propeller. The resulting coefficients are shown in Figure 10 with respect to θ at different V_∞. It is clear that both C_{L_P} and C_{T_P} reduces significantly with the increment in V_∞ and decrement in RPS as the propeller thrust is mainly a function of rotational speed instead of freestream velocity (Figure 8). Hence, the propeller's effect on the WEP system will reduce with the increment in V_∞.

The propeller pitching moment under different θ v.s V_∞ at 625 RPS is shown in Figure 11. As the pitching moment of the propeller is measured off-axis, a correction is made to account for the off-axis loading.

$$M_{x\,Corrected} = M_{x\,Measured} + F_Y\, d_z \tag{3}$$

where d_z is the distance between the propeller and the sensor with respect to the sensor's Z-axis (shown in Figure 4). A positive pitching moment represents a pitch-up moment while a negative pitching moment represents a pitch-down moment. For all incidence angles, with the increment in freestream velocity, $M_x$increases, which also agrees with the trend in our previous study [14]. With the increment in θ, the measured pitch-up moment also increases.

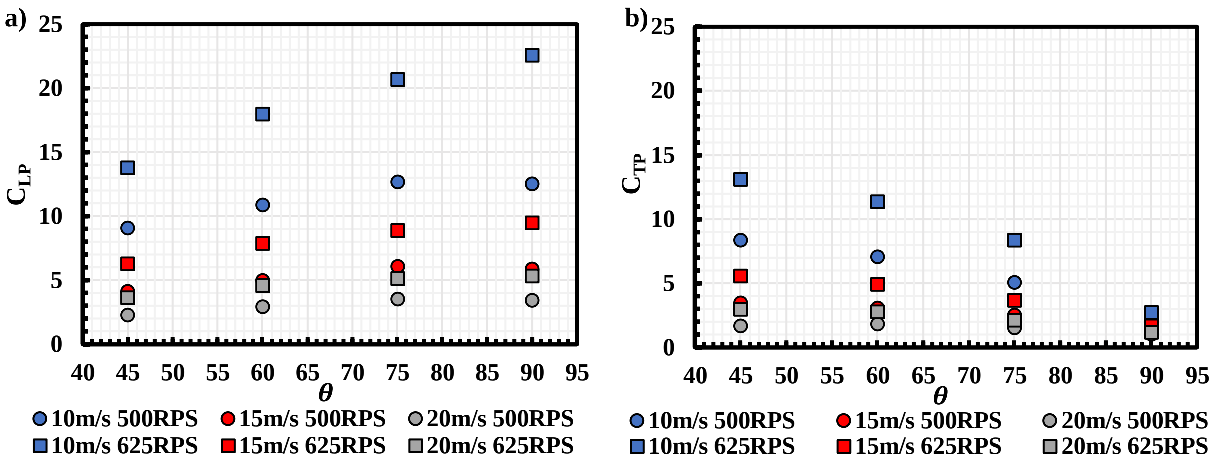

Figure 10. a) $C_{L_P}$and b) C_{T_P} under different V_∞ v.s θ at 500 and 625 RPS

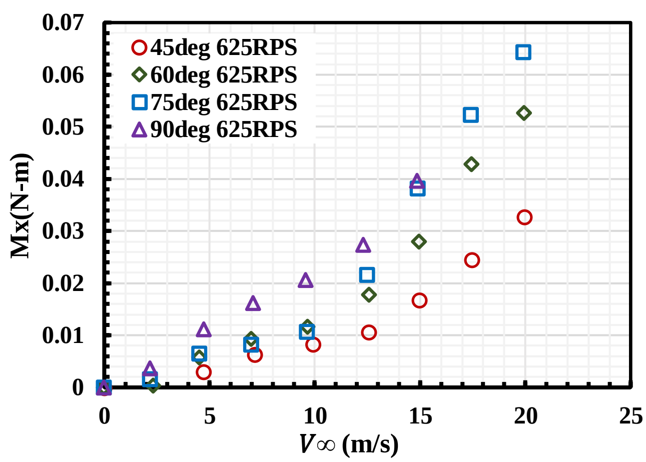

Figure 11. Corrected pitching moment (M_x) of the propeller under different θ v.s V_∞ at 625 RPS Wing Embedded Propeller (WEP) Results

C.Wing Embedded Propeller System

For the wing-embedded propulsion system testing, the propeller rotational speed was held at 625 RPS at three different freestream velocities (10, 15, and 20 m/s) that correspond to three different advance ratios (0.25, 0.38, and 0.51) for the propeller. The results are shown in with respect to the wing reference coordinates, in terms of lift, drag, and pitching moment coefficient.

a) Effect of Propeller on the WEP System

The coefficient of lift from the WEP with the circular hole is shown in Figure 12. The symbols represent different freestream velocities tested and the colors represent the different propeller incidence angles θ in the wing. It is evident that at the same V_∞, with the increment in the propeller mounting incidence angle θ, the magnitude of the lift coefficient at a given α increases due to higher L_P generated at higher incidence angles. The coefficient of lift magnitude decreases with an increase in freestream velocity. While the lift force on the wing scales with the square of the freestream velocity, the L_P from the propeller remains fairly constant with V_∞ as shown in Figure 8. As such, when the net lift from WEP is normalized by a higher velocity, the lift coefficient of the system decreases.

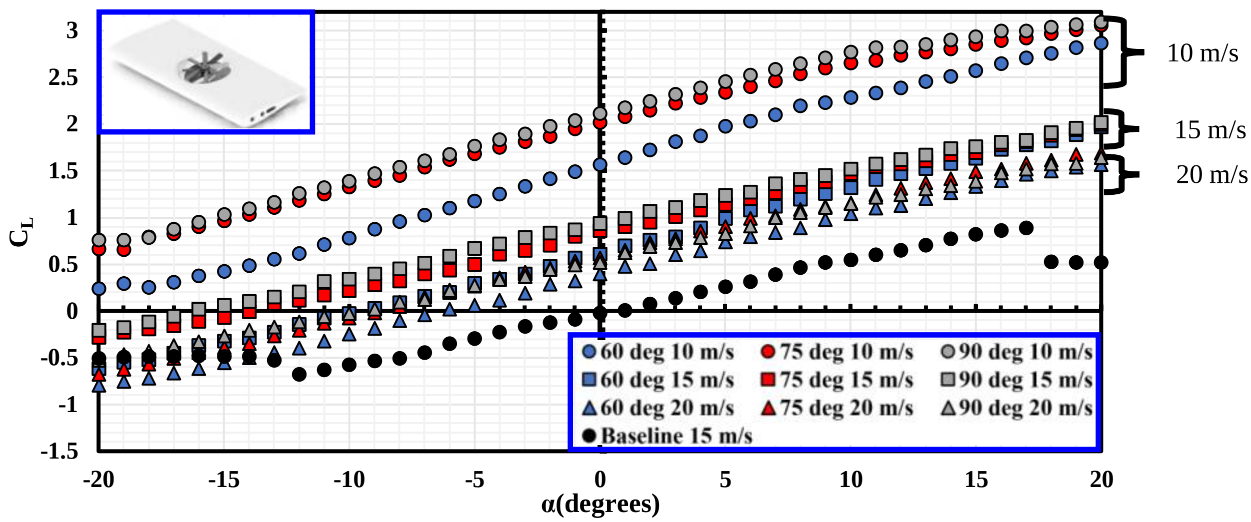

Figure 12: C_L v.s α at various V_∞ and θ_{Mount} for the WEP with a circular hole

The result for the WEP system drags coefficients, C_D, at various V_∞ and θ is shown in Figure 13. As the propeller will generate a forward flight thrust, a negative C_D represents the forward thrust force. At the same V_∞ and α, wing with the propeller mounted at higher inclination angles has a higher effective propeller incidence angle, θ_{Eff}, which can be calculated as:

$$\theta_{Eff} = \theta_{Mount} + \alpha \tag{4}$$

A higher θ_{Eff} results in higher drag force since the forward thrust generation decreases with an increment in inclination angle. With the increment in V_∞, as the variation in T_P generated by the propeller remains relatively constant (Figure 9), the effect of the propeller on the WEP system C_D reduces, similar to C_L results in Figure 12. It is also worth noticing that, at $V_\infty = 20m/s$, the value of C_D is close to or greater than zero for all of the cases tested, indicating the propeller losses its ability to generate the forward thrust for the WEP system.

Figure 14 shows the pitching moment of the WEP system, C_{Mx}, at various V_∞ and θ. Again, a positive C_{Mx} represents a pitch up moment while a negative C_{Mx} represent a pitch down moment. At the lowest V_∞ tested, a positive C_{Mx} is measured for $\theta = 90°$ and 75°, as result of the pitch-up moment generated by the propeller as seen in Figure 11. To balance this pitch-up moment at low flight speed, an additional pitch-down moment would be required from the empennage of the aerial vehicle. However, with the increment in V_∞, the effect of the propeller on the overall WEP system reduces, C_{Mx} of the WEP system becomes negative as due to the natural pitch-down moment of the wing.

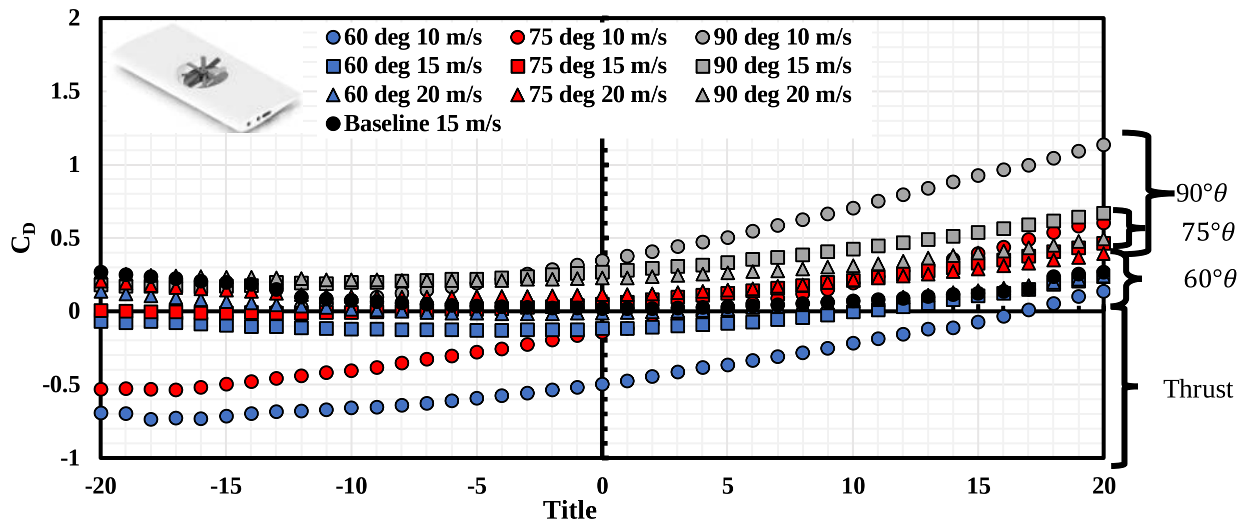

Figure 13. C_D v.s α at various V_∞ and θ_{Mount} for the WEP with a circular hole

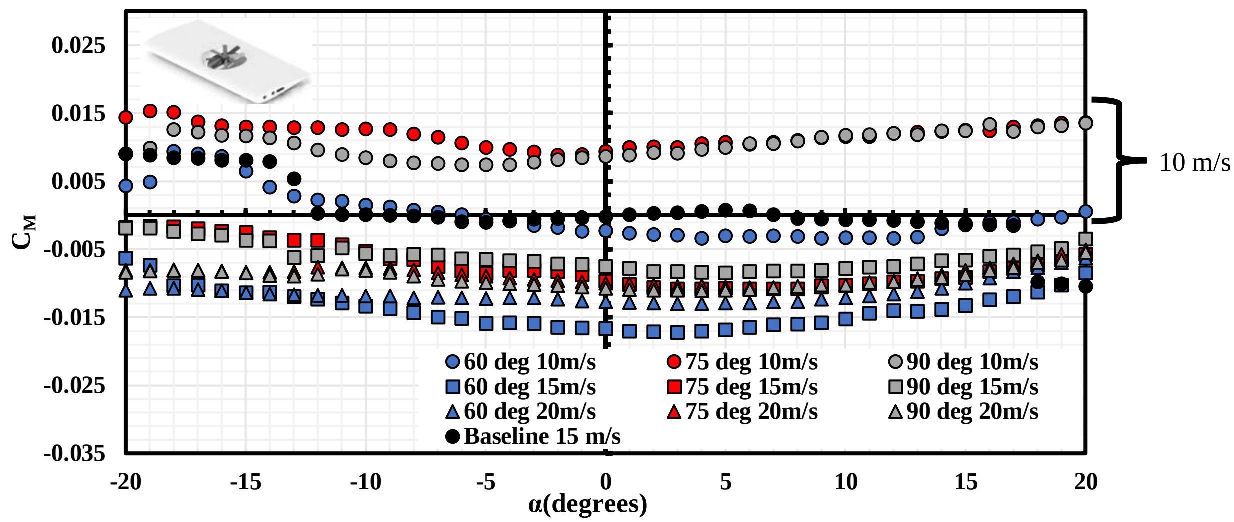

Figure 14: C_M v.s α at various V_∞ and θ_{Mount} for the WEP with a circular hole

b) Effect of Wing Shape on the WEP System

As discussed previously, three different wing hole shapes are investigated in the study. To understand the effect of the hole shape on the WEP system performance, the C_L v.s α for all three wing hole cases with various θ_{Mount} at V_∞ of 10 m/s is shown in Figure 15. Differing from the plots in the last section, the same color legends indicate the same wing hole while the same symbol legends indicate the same θ_{Mount}. At $\theta_{Mount} = 60°$, the circular wing hole produces a much higher lift coefficient than both square wing holes, resulting in a lower propeller-wing interference, as the difference in lift coefficient between the standalone wing is relatively small, as seen in Figure 7. A lower propeller-wing interference for the circular wing hole is also observed in our previous study [5]. For the square wing hole, adding the fillet to smooth out the edges results in a better performance of the WEP system, though still being outperformed by the circular wing hole. With the increment in θ_{Mount}, the trends seen in Figure 10 is preserved.

At a higher V_∞ of 20 m/s (Figure 16), the overall C_L magnitude and the difference in C_L between wing hole cases reduce due to the reduction of the overall propeller effect on the WEP system at large V_∞, as discussed previously in Figure 12. Again, the trend follows the one discussed in Figure 15, with a lower θ_{Mount} showing a higher C_L for the circular wing hole when compared to other cases. Similarly, the difference in C_L also reduces with the increment in θ_{Mount}, with the fillet square wing hole performs almost the same as the circular wing hole.

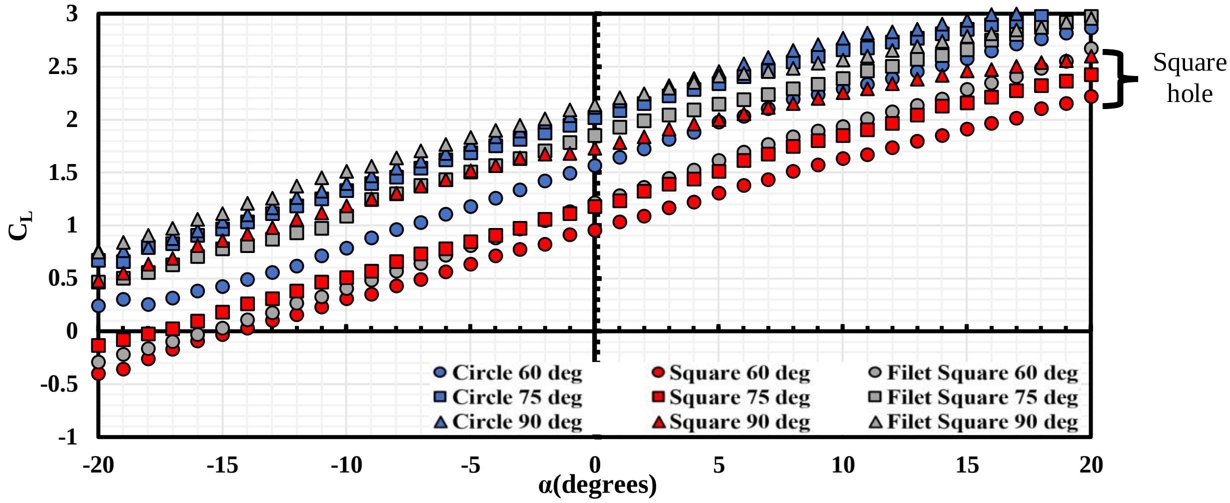

Figure 15. C_L v.s α for different wing hole shapes with various θ_{Mount} at $v_\infty = 10\ m/s$

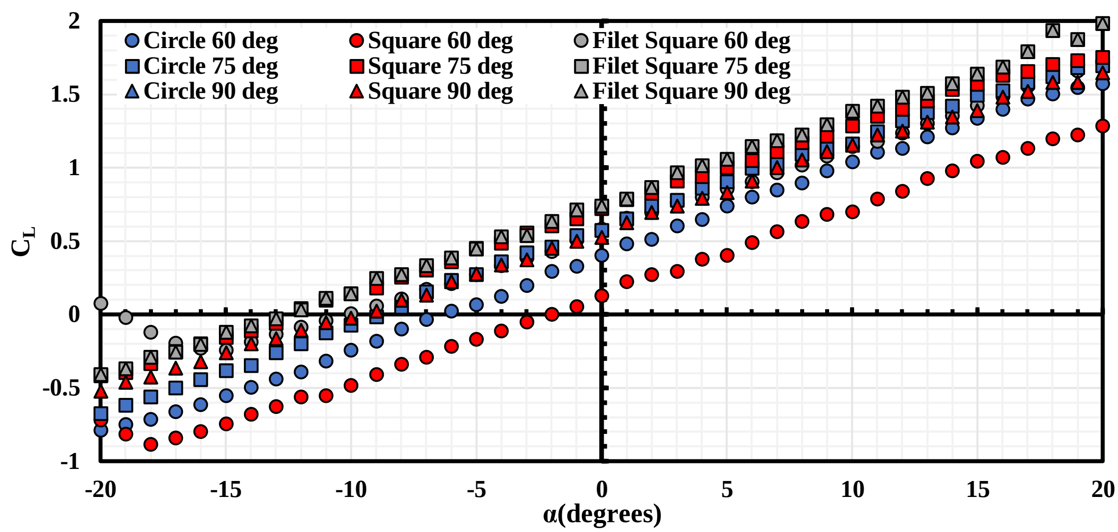

Figure 16. C_L v.s α for different wing hole shapes with various θ_{Mount} at $v_\infty = 20\ m/s$

Figure 17 shows C_D v.s α for the cases shown in Figure 15. Again, a positive C_D indicates the drag force experienced by WEP system while a negative C_D represents a forward thrust force generated by the WEP system. Similar to the results seen in Figure 13, θ_{eff} reduces with the decrement α resulting a higher forward thrust generation for all θ_{Mount} cases. At $V_\infty = 10m/s$ and $\theta_{Mount} = 60°$, the square hole wing has a much higher C_D when compared to the circular and fillet square wing hole, indicating a much lower forward thrust generation ability and a higher propeller-wing interference. Again, adding the fillet to smooth the sharp edges of the square wing results in a much better performance in C_D as all fillet square wing cases outperforms the square wing hole. Surprisingly, the fillet square hole was able to generate a slightly higher forward thrust at $\alpha < 10°$ despite a worse performance in the lift generation. With increment in θ_{Mount}, the benefit of the circular wing hole becomes significant, with a better thrust generation at $\theta_{Mount} = 75°$ and lower drag at $\theta_{Mount} = 90°$.

At higher V_∞ of 20 m/s shown in Figure 18, due to the increment in the drag force on the wing, while the forward thrust generated by the propeller remains relatively constant, as discussed in Figure 13, only the circular and fillet square wing hole cases were able to generate a slight forward thrust at $\theta_{Mount} = 60°$. Again, for all θ_{Mount} cases, the square wing hole has the worst performance in forward thrust generation among all cases. On the other hand, the fillet square wing was able to produce higher forward thrust when compared to the circular wing at $\theta_{Mount} = 60°$ and $75°$, which is likely a result of a higher propeller intake flow area due to a higher hole area for the fillet square wing.

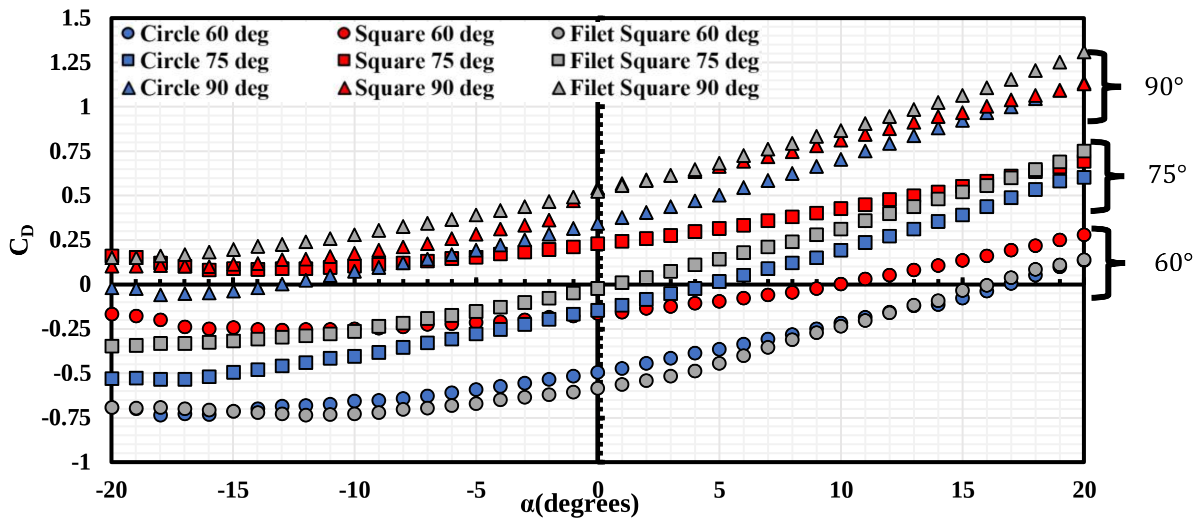

Figure 17. C_D v.s α for different wing hole shapes with various θ_{Mount} at $v_\infty = 10\ m/s$

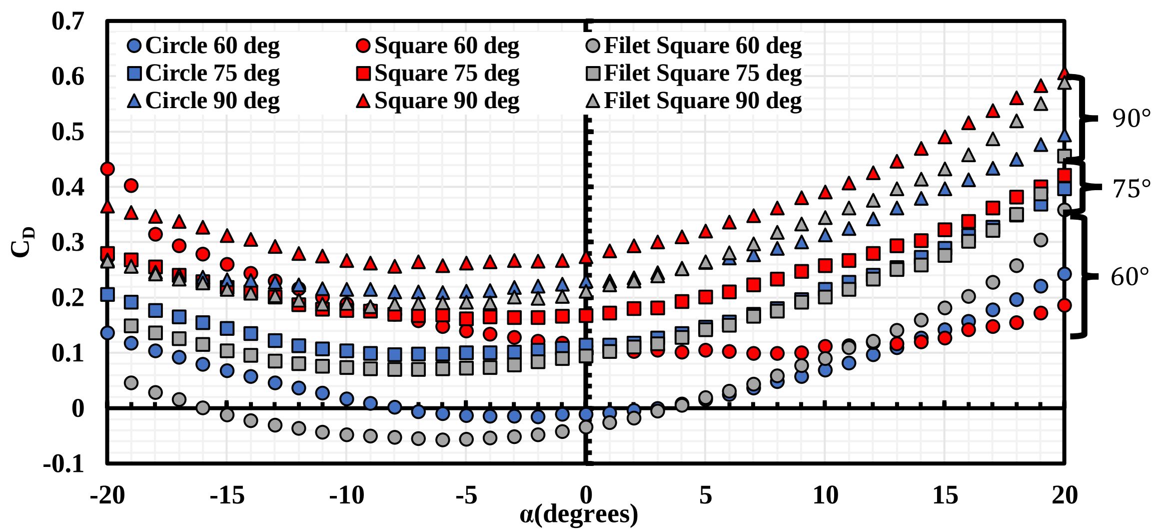

Figure 18: C_D v.s α for different wing hole shapes with various θ_{Mount} at $v_\infty = 20\ m/s$

Figure 19 shows the WEP system's pitching moment coefficient, C_M for different hole shapes. Like the trend observed in Figure 14, due to the pitch-up moment of the propeller at high θ_{eff} as well as the large influence of the propeller on the WEP system at low V_∞, a pitch up moment of the WEP system is observed for most of the cases. At $\theta_{Mount} = 60°$, the difference between different hole shape on C_M is relatively small, with the increment in θ_{Mount}, C_M of the WEP system increases, as well as the difference of C_M between different wing holes. The square wing hole has the lowest C_M at large θ_{Mount} because of its poor performance in forward thrust and lift generation seen in Figure 15 and Figure 17. For the circular wing hole, the value of C_M is lower than the fillet square wing hole even though its performance in forward thrust and lift generation is better, indicating better stability for the circular wing hole at low V_∞.

With the increment in V_∞, the value of C_M becomes more negative because of the wing's natural pitch-down moment and the increment in the wind influence on the WEP system at high V_∞. The effect of θ_{Mount} also reduces. Like the $V_\infty = 10m/s$ case, the square wing hole has a relatively low C_M due to the lower forward thrust and lift generation, while the circular wing hole has a much lower C_M when compared to the fillet square hole indicating its better performance on the stability of the WEP system.

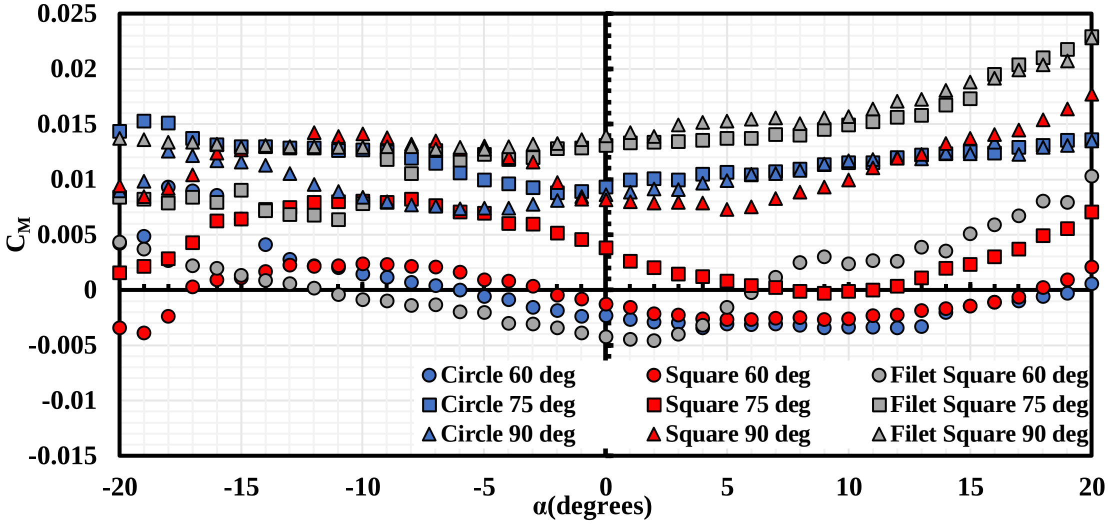

Figure 19: C_M v.s α for different wing hole with various θ_{Mount} at $v_\infty = 10\ m/s$

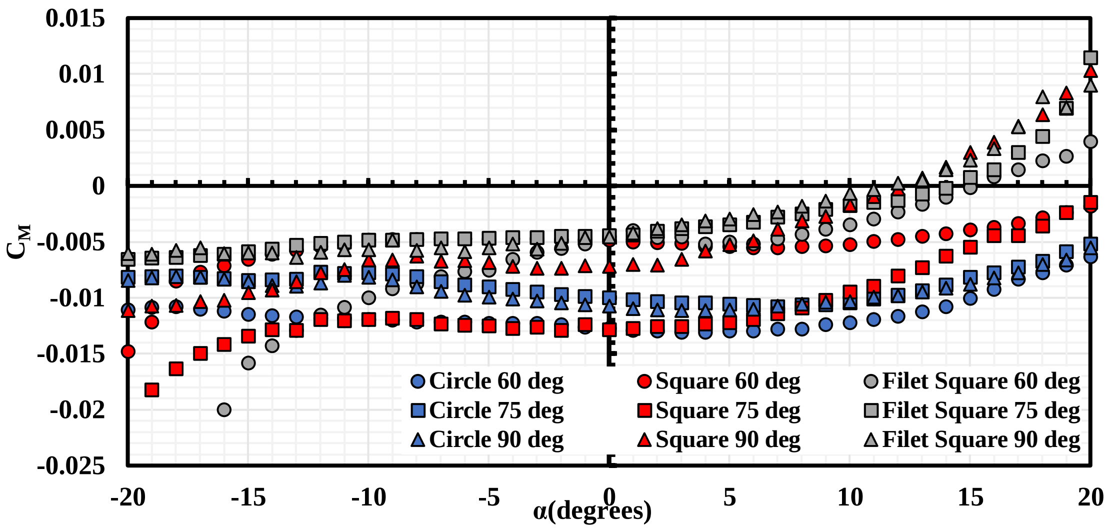

Figure 20: C_M v.s α for different wing hole with various θ_{Mount} at $v_\infty = 20\ m/s$

D. Overall Efficacy of the Wing Embedded Propeller System

To extract the interference effects between the propeller and the wing in the WEP system and to determine the overall efficacy of the WEP system, the individual performance of the standalone wing (Figure 7) and standalone propeller (Figure 8) are summed to calculate the ideal performance of the WEP system. For the standalone propeller performance, a second-order interpolation is conducted at each V_∞ among all θ cases tested to get the performance at any intermediate θ cases at this very specific V_∞. The ideal lift (L_{ideal}) and ideal drag (D_{Ideal}) is calculated as:

$$L_{ideal} = L_P + L_{Wing} \qquad D_{ideal} = T_P + D_{Wing} \tag{5}$$

The lift and drag coefficient of the idea force is then calculated accordingly:

$$C_{L_{Ideal}} = \frac{L_P + L_{Wing}}{qS_{Ref}} = C_{L_P}\frac{S_{Prop}}{S_{wing}} + C_{L_{Wing}} \qquad C_{D_{Ideal}} = \frac{D_P + D_{Wing}}{qS_{Ref}} = C_{D_P}\frac{S_{Prop}}{S_{wing}} + C_{D_{Wing}} \tag{6}$$

The coefficients are then compared with the measured WEP system performance discussed in the previous sections. As the circular wing hole shows the best performance among all wing hole cases with a higher overall lift and forward thrust generation. The comparison of C_L for $\theta_{Mount} = 60°$ and 90° is shown below in Figure 21. The overall trend of C_L for the measured and ideal conditions agrees, except for extreme α conditions where the stall occurs on the standalone wing. Therefore, the existence of minimal interference at lower angles of attack allows for fairly accurate prediction of WEP C_L through linear addition of the standalone wing and propeller performance. For both θ_{Mount} cases, the difference between the ideal and measured C_L increases at negative α, indicating a higher propeller-wing interference at this region. This difference also increases when reducing the θ_{Mount} due to a higher propeller-wing interference.

The difference between the $C_{L_{WEP}}$ and $C_{L_{ideal}}$, ΔC_L and the difference between $C_{D_{WEP}}$ and $C_{D_{ideal}}$, ΔC_D, is then calculated. Figure 22 shows ΔC_L and $\Delta C_{L_{Norm}}$ for the circular wing hole at $\theta_{Mount} = 60°$. Since a large ΔC_L is observed at stall α, the discussion will focus on $-12° < \alpha < 12°$ for practical applications. As seen in Figure 21, a higher ΔC_L is observed at negative α and lower V_∞, ΔC_L reduces with the increment of α, indicating a lower wing-propeller interference, which is also reflected in Figure 22 a. This is potentially due to the prolonged flow attachment caused by the suction effect of the propeller on the upper surface of the wing with increase in α, hence acting like a blown wing configuration.

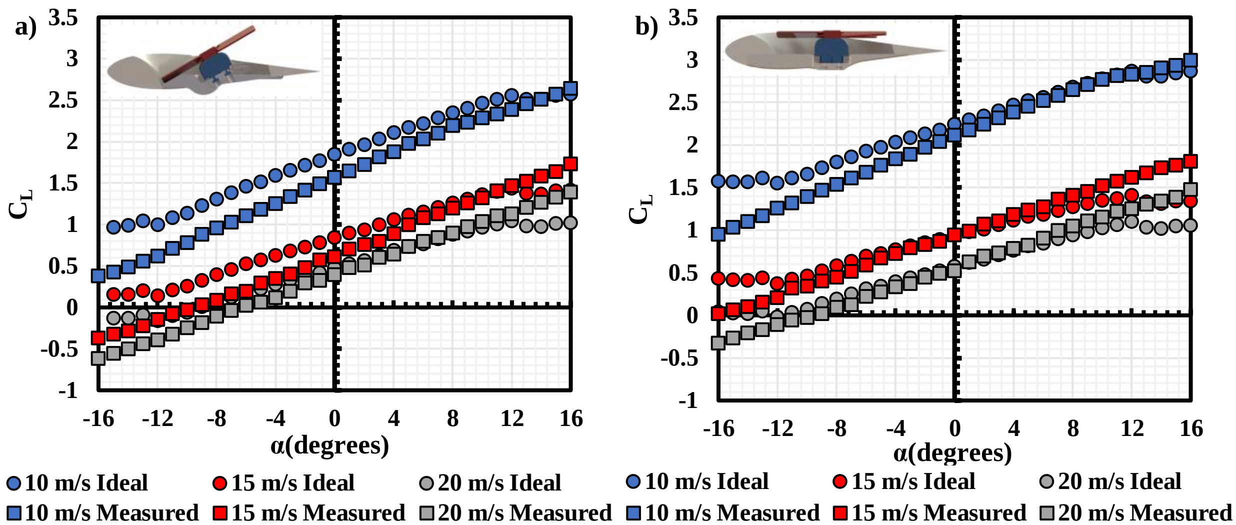

Figure 21: Measured and ideal C_L for the circular wing hole WEP system at various V_∞ for a) $\theta_{Mount} = 60°$ and b) $\theta_{Mount} = 90°$.

However, ΔC_L does not accurately represent the losses in the WEP performance as the magnitude of C_{L_P} reduces at higher V_∞ as discussed in Figure 10. Hence, a relatively higher ΔC_L will be calculated at lower V_∞. Hence, to eliminate such effects, $C_{T_{total}}$ at the same V_∞ is used to normalize ΔC_L and ΔC_D with respect to the WEP reference area, S_{ref}, as:

$$\Delta C_{L_{Norm}} = \frac{C_{L_{WEP}} - C_{L_{Ideal}}}{C_{T_{Total}}(S_{Prop}/S_{Ref})} * 100 \qquad \Delta C_{D_{Norm}} = \frac{C_{D_{WEP}} - C_{D_{Ideal}}}{C_{T_{Total}}(S_{Prop}/S_{Ref})} * 100 \tag{7}$$

If one assumes the performance of the wing remains constant, $\Delta C_{L_{Norm}}$ and $\Delta C_{D_{Norm}}$ then represents the changes in the propeller performance when installed on the WEP wing with a hole, and accounts for all wing-propeller interference as the changes in $\Delta C_{L_{Norm}}$ and $\Delta C_{D_{Norm}}$. A positive $\Delta C_{L_{Norm}}$ represents a higher lift measured for the WEP system when compared to the ideal condition while a positive $\Delta C_{D_{Norm}}$ represents a higher forward thrust measured. $\Delta C_{L_{Norm}}$ for the circular wing hole at $\theta_{Mount} = 60°$ is shown in Figure 22 b. The general trend follows that of Figure 22 a, where $\Delta C_{L_{Norm}}$ reduces with the increment of α it is worth noticing that at $V_\infty = 10 m/s$ ($J = 0.25$ for the propeller), the magnitude of ΔC_L is the highest among all V_∞ cases, while the magnitude for $\Delta C_{L_{Norm}}$ is the lowest, except for $\alpha > 6°$, indicating an overall lower performance loss in propeller lift generation at lower V_∞, with an average of -10% of $\Delta C_{L_{Norm}}$ at positive α range. Moreover, at $V_\infty = 20 m/s$ ($J = 0.51$ for the propeller), a positive

$\Delta C_{L_{Norm}}$ is observed at $\alpha > 4°$ indicating better WEP system performance than the ideal case, as a result of the prolonged flow attachment caused by the suction effect of the propeller on the upper surface of the wing.

ΔC_L and $\Delta C_{L_{Norm}}$ for $\theta_{Mount} = 75°$ and $90°$ is shown in Figure 23. Similarly, ΔC_L and $\Delta C_{L_{Norm}}$ increases with the increment in α for all θ_{Mount} cases, indicating a better performance of the WEP system at positive α. For $V_\infty = 10m/s$ ($J = 0.25$), $\Delta C_{L_{Norm}}$ is less than 10% at negative α and almost negligible at positive α for both θ_{Mount} cases. Again, with the increment in V_∞, the magnitude of $\Delta C_{L_{Norm}}$ increases at negative α. At $20m/s$ ($J = 0.51$), the benefit of the propeller on the wing lift force also increases when compared to $\theta_{Mount} = 60°$ case, especially for $\theta_{Mount} = 75°$ case as the circular wing hole shape is designed for a $\theta_{Mount} = 75°$ condition. Hence, when placing the propeller in the WEP system, the performance of the WEP system in terms of lift generation at positive α can be directly estimated by the summation of the performance of the standalone wing and standalone propeller (Equation 8), which can be estimated with medium-fidelity CFD simulation or traditional analytical methods.

$$C_{L_{WEP}} = C_{L_{Wing}} + \frac{T_P + T_{total}\,\Delta C_{L_{Norm}}}{qS_{ref}} \tag{8}$$

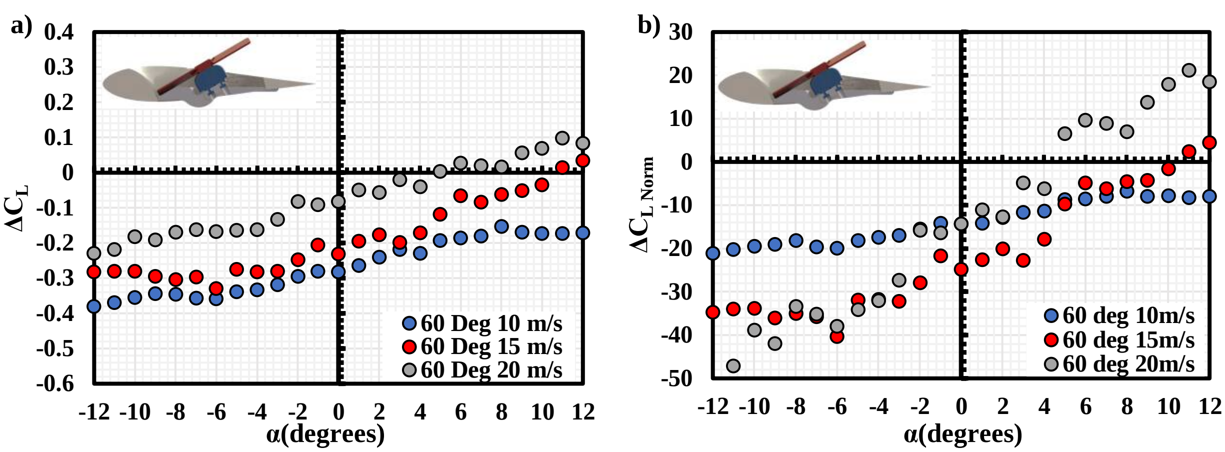

Figure 22: a) ΔC_L and b) $\Delta C_{L_{Norm}}$ for circular hole wing at various V_∞ and $\theta_{Mount} = 60°$

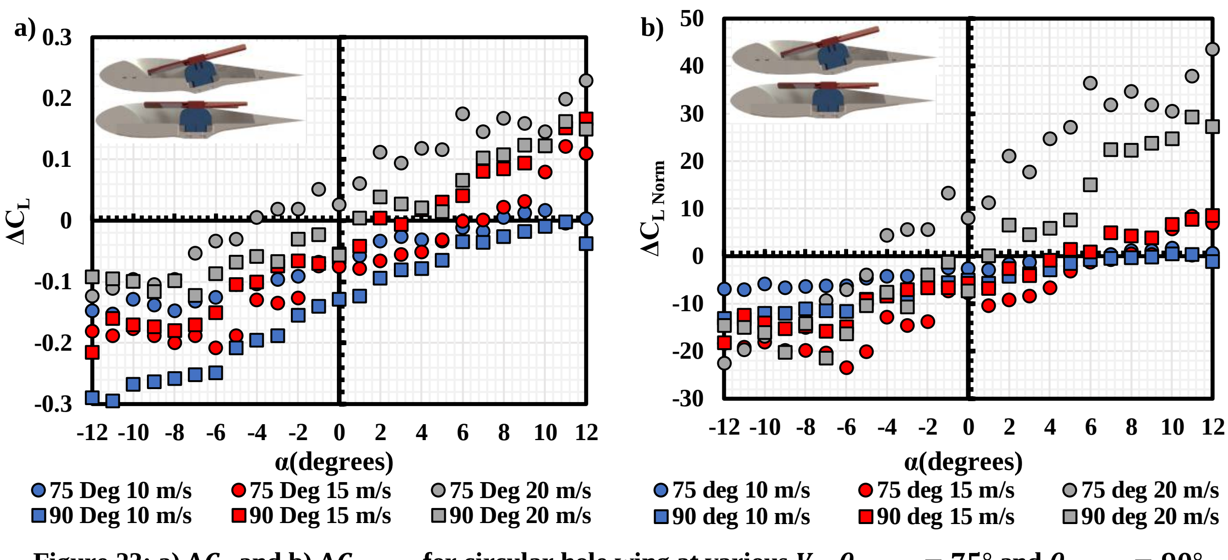

Figure 23: a) ΔC_L and b) $\Delta C_{L_{Norm}}$ for circular hole wing at various V_∞ $\theta_{Mount} = 75°$ and $\theta_{Mount} = 90°$

On the other hand, the ideal and measured C_D, as well as ΔC_D, for the circular wing hole at $\theta_{Mount} = 60°$ and $75°$ is shown in Figure 24. The $\theta_{Mount} = 90°$ is not discussed in terms of C_D as it is not practical for forward thrust generation conditions. The difference between the ideal and measured C_D is more drastic when compared to the difference in C_L, especially at negative α where the propeller-wing interference is also significant for the lift generation, indicating a more significant loss in the forward thrust generation of the WEP system. Moreover, with the increment in θ_{Mount}, ΔC_D stays relatively constant at the same V_∞ and propeller advance ratio.

Again, to better quantify the performance loss on the propeller forward thrust generation, $\Delta C_{D_{Norm}}$ is calculated (Equation 8), and the result is shown in Figure 25. ~30% loss in the forward thrust generation for the propeller is observed at $V_\infty = 10m/s$ ($J = 0.25$) for both θ_{Mount} cases and stays relatively constant across the low α angels. At higher forward flight speed, $\Delta C_{D_{Norm}}$ increases to ~40% for $V_\infty = 15m/s$ ($J = 0.38$) and ~50% for $V_\infty = 20m/s$ ($J = 0.51$) cases on average. This is likely a result of large flow separation occurring at the lower surface of the WEP system due to the wake of the propeller, this flow separation increases the overall drag coefficient of the wing, and is being magnified at large V_∞ as the influence of the wing on the WEP system increases. Moreover, $\Delta C_{D_{Norm}}$ reduces with the increment in α, indicating a lower propeller-wing interference at the positive α range. Overall, the WEP system suffers a strong loss in the forward flight thrust generation due to the propeller-wing interaction. The summation of two standalone wings and propellers is not applicable for the estimation of the forward thrust generation, and further analysis is required during the system design process.

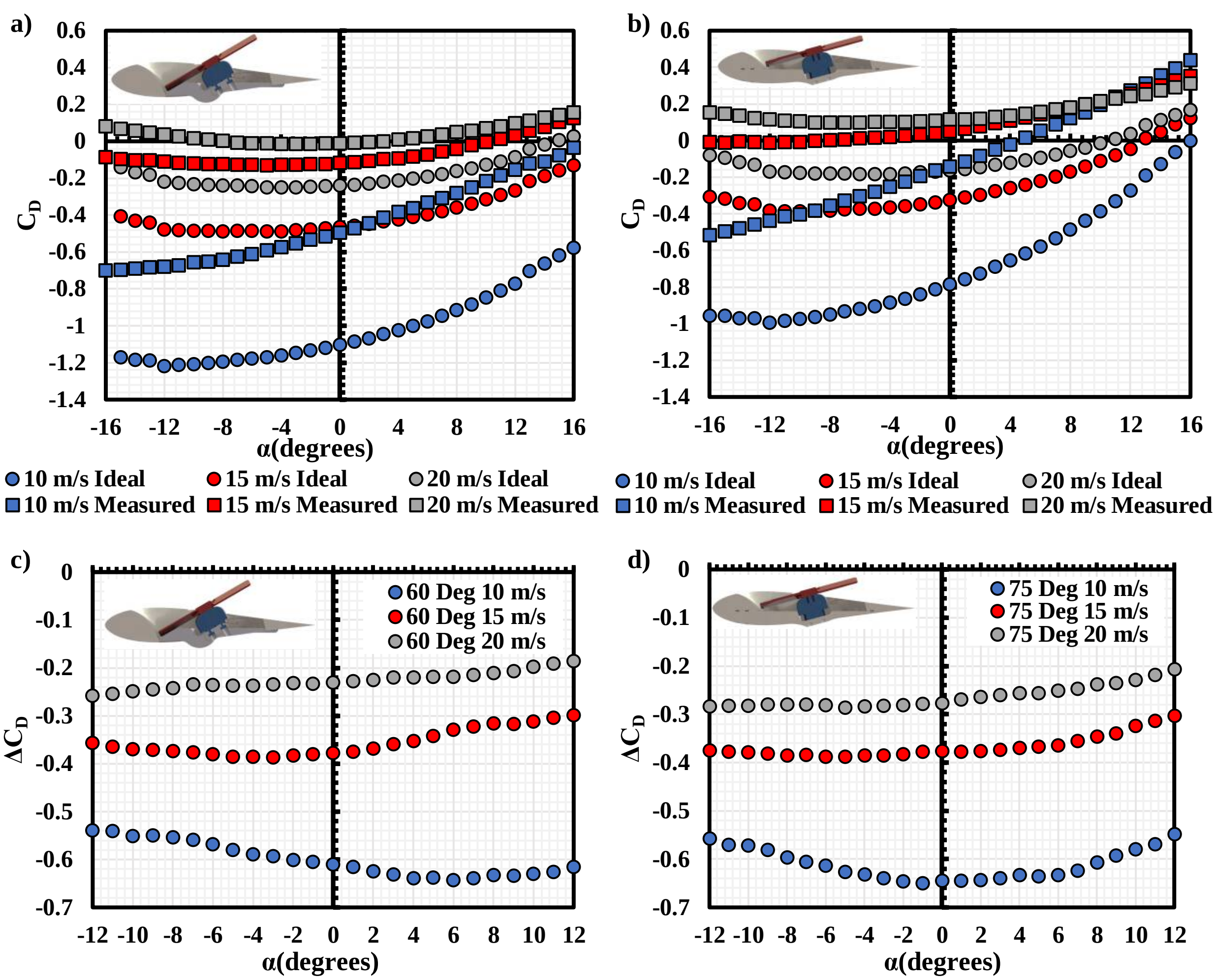

Figure 24. Measured and ideal C_D for a) $\theta_{Mount} = 60°$ and b) $\theta_{Mount} = 75°$ along with ΔC_D for c) $\theta_{Mount} = 60°$ and d) $\theta_{Mount} = 75°$ for the circular wing hole WEP system at various V_∞

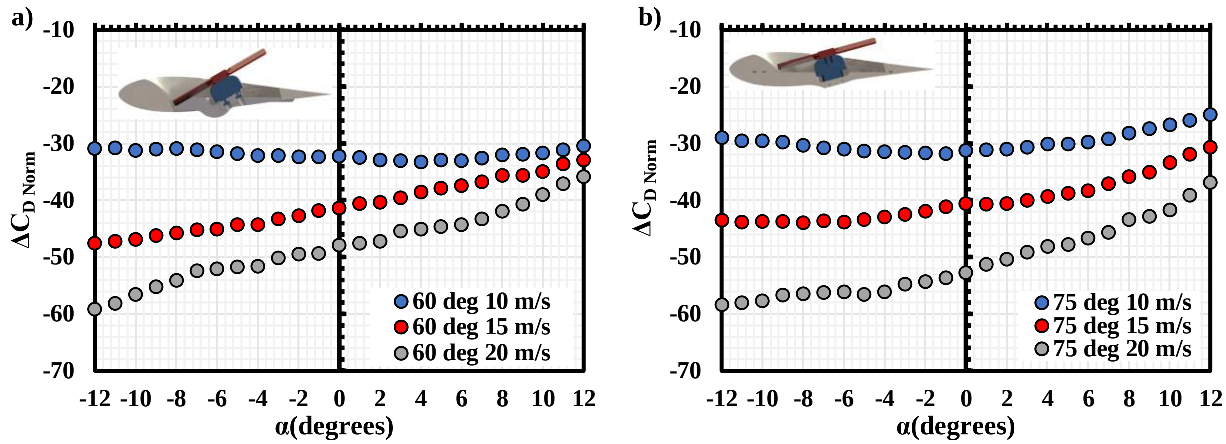

Figure 25: **ΔF_S for circular wing hole at various V_∞ with a) $\theta_{Mount} = 60°$ and b) $\theta_{Mount} = 75°$**

IV.Conclusion

The Wing Embedded Propeller (WEP) system was experimentally investigated as one system in the study using three different wing hole shapes. For the standalone wings with holes, the presence of the different shaped holes didn't yield measurable differences among each other. However, the lift curve slope was lower when compared to the baseline wing with no hole suggesting a decrease in the effective AR of the wing.

The wing hole shape has a higher effect on the WEP performance when the propeller is installed. The circular wing hole has the best overall performance in the lift and forward flight thrust generation among all wing hole shape cases. The propeller influence on the WEP system reduces while the influence of the wing on the WEP system increase with the increment in forward flight speed, as the propeller thrust does not change quadratically with the freestream velocity, while the forces on the wing do.

The performance of the WEP lift generation can be estimated by the summation of the standalone propeller and wing performance at positive α, as the propeller-wing interference is almost negligible. Moreover, at large forward flight speed, a benefit of the propeller on the WEP lift generation is also observed.

However, such a summation of the standalone performance is not applicable to the forward thrust estimation of the WEP system. A significant loss in forward thrust generation is observed in all cases, which also increases with the increment of forward flight speed, resulting in a loss of the forward thrust generation of up to 50% at the highest forward flight speed tested.

V.Acknowledgments

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